



OBJECT ORIENTED ANALYSIS AND DESIGN

Ch - 4

Use Case Modeling

A study material for the students of GLS University

What is Use Case

A use case tells a complete story of how a user interacts with a system to achieve something useful.

- "Use" = someone doing something
- "Case" = a situation or goal

So, a Use Case = A situation where the system is used to achieve a goal.

Use Cases are not about how the system works inside, but what the user wants to do and how the system responds.

Use cases help in:

- Understanding what the system should do
- Describing user goals
- Planning features in software
- Creating test cases later

A Test Case is a set of steps used to check if a feature of the system is working correctly.

Example

A use case diagram is a visual representation that shows -

- Who uses the system (Actor)
- What they do (Use Cases)
- The System Boundary

Actor	Use Case (What they want to do)
Customer	Register
Customer	Login
Customer	Search Products
Customer	Add to Cart
Customer	Make Payment
Admin	Add Products
Admin	View Orders

Use Case Modeling

Use Case Modeling is a simple way of understanding and writing down what the user wants from a software system.
It helps us answer:

- “Who will use the system?”
- “What will they do with it?”

It is part of requirements engineering — that is, figuring out what the software should do before we start building it.

Helps developers and clients understand the system clearly

Focuses on user goals instead of technical details

Acts as a blueprint before actual design and coding

Components of Use Case Modeling

1. System Boundary: It is a box that shows what is part of the system and what is not.

Everything inside the box = system features

Everything outside the box = external users/systems

2. Actors: An actor is anyone or anything outside the system that interacts with it.

Examples: User, Admin, External systems (e.g., Payment Gateway)

3. Use Cases: These are the goals or tasks that actors want to achieve using the system.

Examples: Login, Register, Buy Product, Generate Report

4. Relationships: These are the connections between actors and use cases or between use cases themselves.

Finding Actors

The actors of a system can be identified by answering a number of questions:

- Who will use the functionality of the system?
- Who will maintain the system?
- What devices does the system need to handle?
- What other system does this system need to interact?
- Who or what has interest in the results of this system?

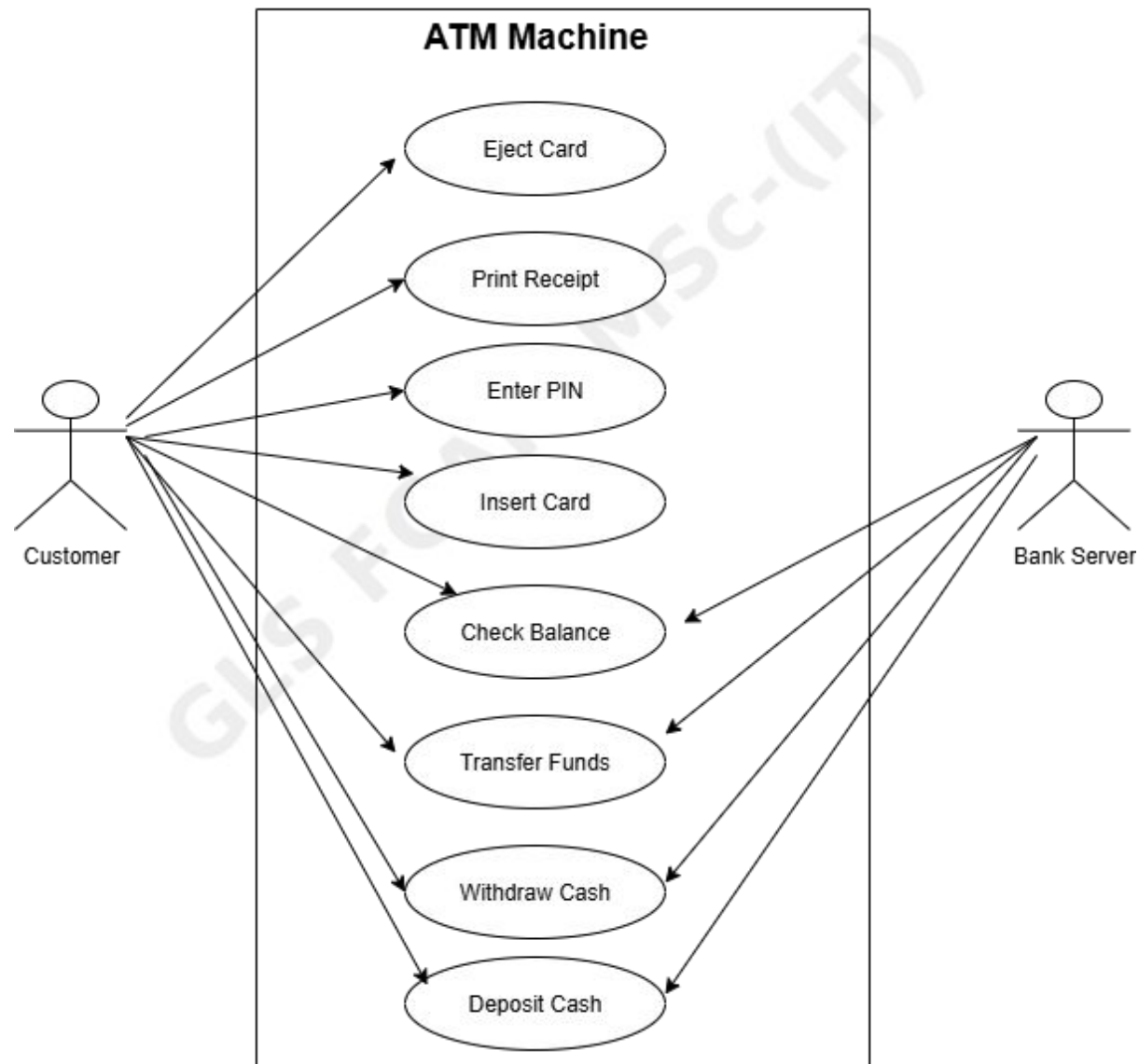
Finding Use-cases

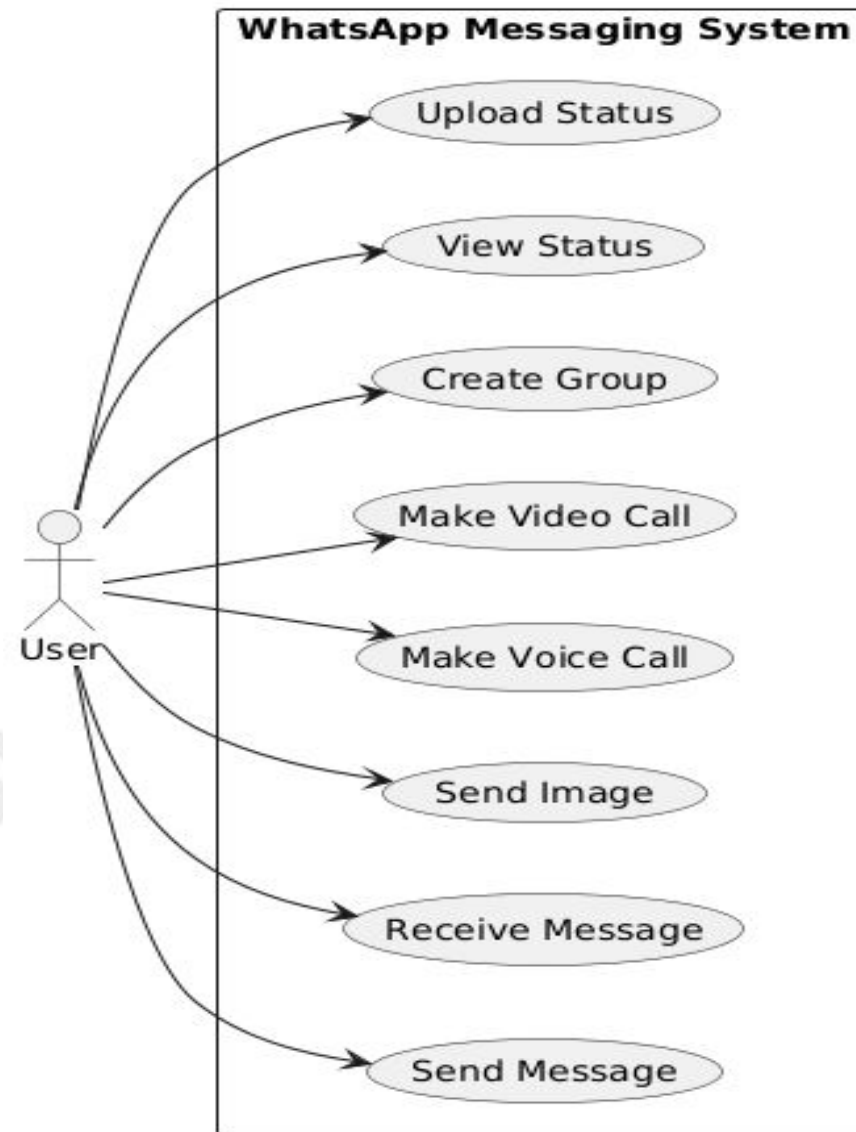
For each actor ask these questions:

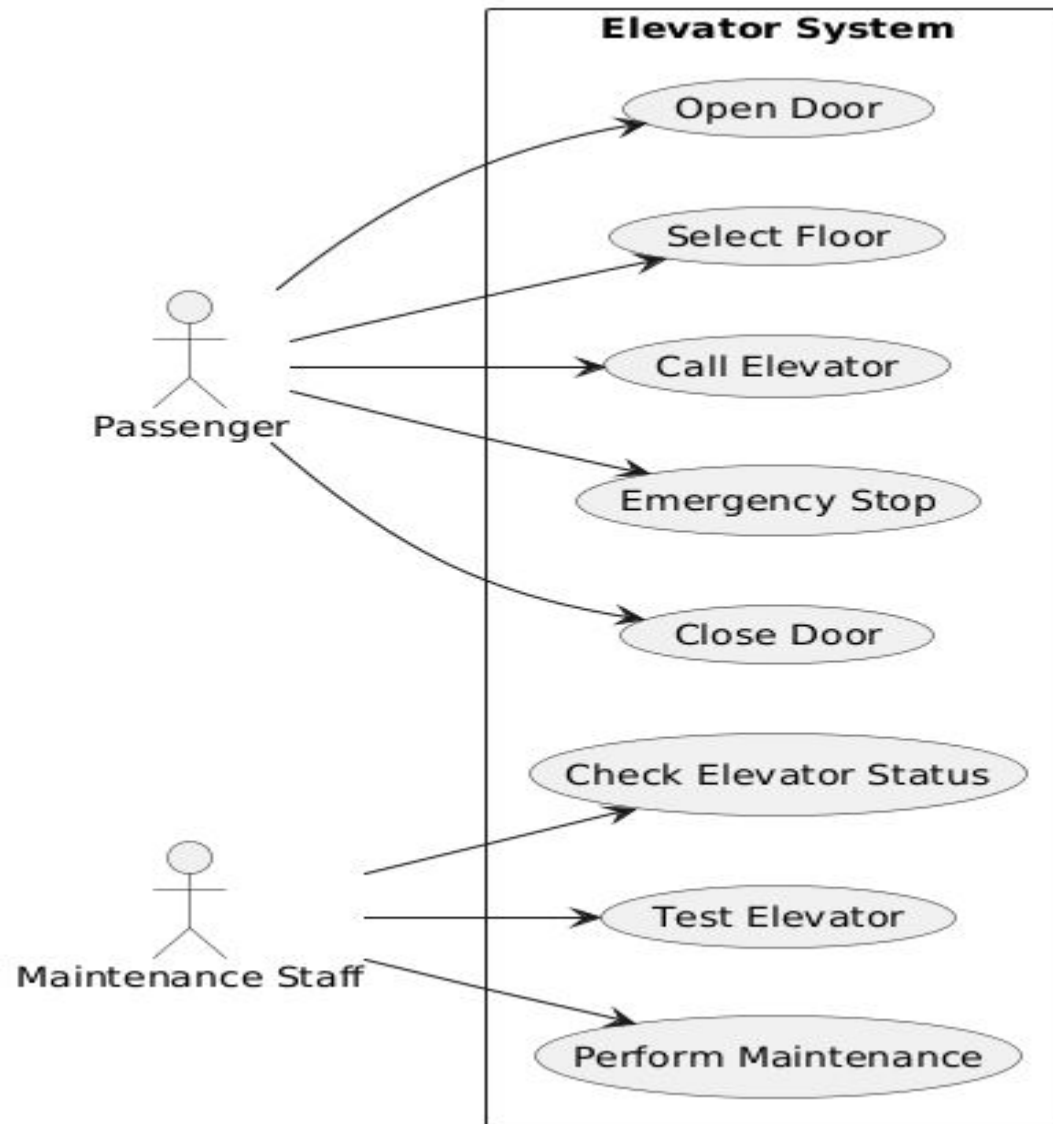
- Which functions does the actor require from the system?
- What does the actor need to do?
- Could the actor's work be simplified or made efficient by new functions in the system?
- What events are needed in the system?
- What are the problems with the existing systems?
- What are the inputs and outputs of the system?

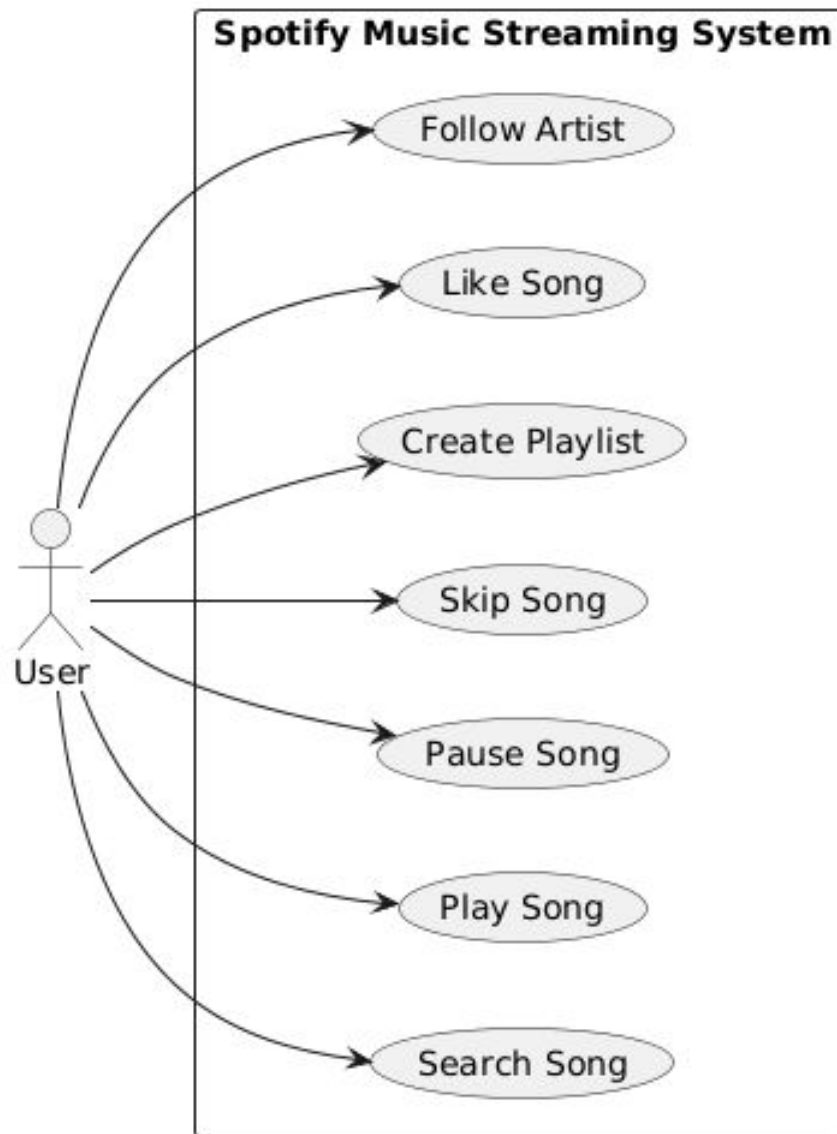
Use Case Diagram – Basic Symbols

Symbol	Meaning
 Stick Figure	Actor (a person or system interacting with your software)
 oval	Use Case (function or service the system provides)
 Rectangle with name	System Boundary (defines what is inside/outside the system)
 Line	Association (connection between actor and use case)
 Arrow with <<include>>	Include Relationship (a common sub-use case is always included)
 Arrow with <<extend>>	Extend Relationship (optional/conditional behavior)









Task

1. Online Movie Ticket Booking: A user wants to search for movies, select a showtime, book seats, and make a payment online.

Task: Identify actors and use cases. Draw a use case diagram.

2. Ride-Hailing App (like Uber): A rider books a ride, tracks driver location, and makes payment. A driver accepts rides and completes trips.

Task: List actors and their associated use cases.

3. YouTube-like Video Platform: Users sign up, watch videos, like/share content, and upload their own videos. Admin reviews flagged videos.

Task: Identify 2 types of actors and their use cases.

Types of Actors

1. Primary Actor: The actor who initiates the interaction with the system to achieve a specific goal.

Goal-Oriented: They use the system to fulfill a business or user goal.

Examples:

Customer using an ATM to withdraw money

User playing music on Spotify

2. Secondary Actor (Supporting Actor): The actor that supports the system in carrying out the use case. They do not initiate the use case.

Role: Provide a service or functionality needed by the system to help the primary actor achieve their goal.

Examples:

Bank server verifying ATM transactions

Payment gateway in online shopping

Case Study: The NextGen POS System

What is a POS System?

POS = Point of Sale

Used at billing counters in stores Combines hardware + software

Scans products, calculates bill, processes payments

Why NextGen POS?

- Replace old cash register systems
- Modern, flexible, and scalable
- Works for all types of retail stores
- Real-time sales and inventory integration

Key Business Goals

- Modernization
- Flexibility across stores
- Real-time inventory updates
- Better customer service
- Secure and fast transactions

Actors in the System

1. Cashier -

The cashier is the primary actor.

The cashier uses the POS system to process customer purchases.

The cashier can:

- Start a sale
- Scan product
- Enter quantity
- Remove product
- Calculate total
- Process payment
- Generate receipt

Actors in the System

2. Store Manager -

The manager manages products, inventory, and sales information.

The manager can:

- Add product
- Update product
- Remove product
- Check inventory
- View sales
- Generate reports

Actors in the System

3. Customer -

The customer interacts indirectly with the POS system during a purchase.

For example:

- Provides products for purchase
- Makes payment
- Receives receipt

4. Payment Gateway -

The payment gateway is an external system.

It interacts with the POS system when the customer pays using: Card, UPI, Digital payment

5. Inventory System -

If the store has a separate inventory system, it can interact with the POS system to update stock.

Main Use Cases

Cashier -

The Cashier can:

- Start Sale
- Scan Product
- Enter Product
- View Product Details
- Update Quantity
- Remove Product
- Calculate Total
- Process Payment
- Generate Receipt
- Cancel Sale

Store Manager -

The Manager can:

- Add Product
- Update Product
- Remove Product
- View Product
- Check Inventory
- View Sales
- Generate Sales Report

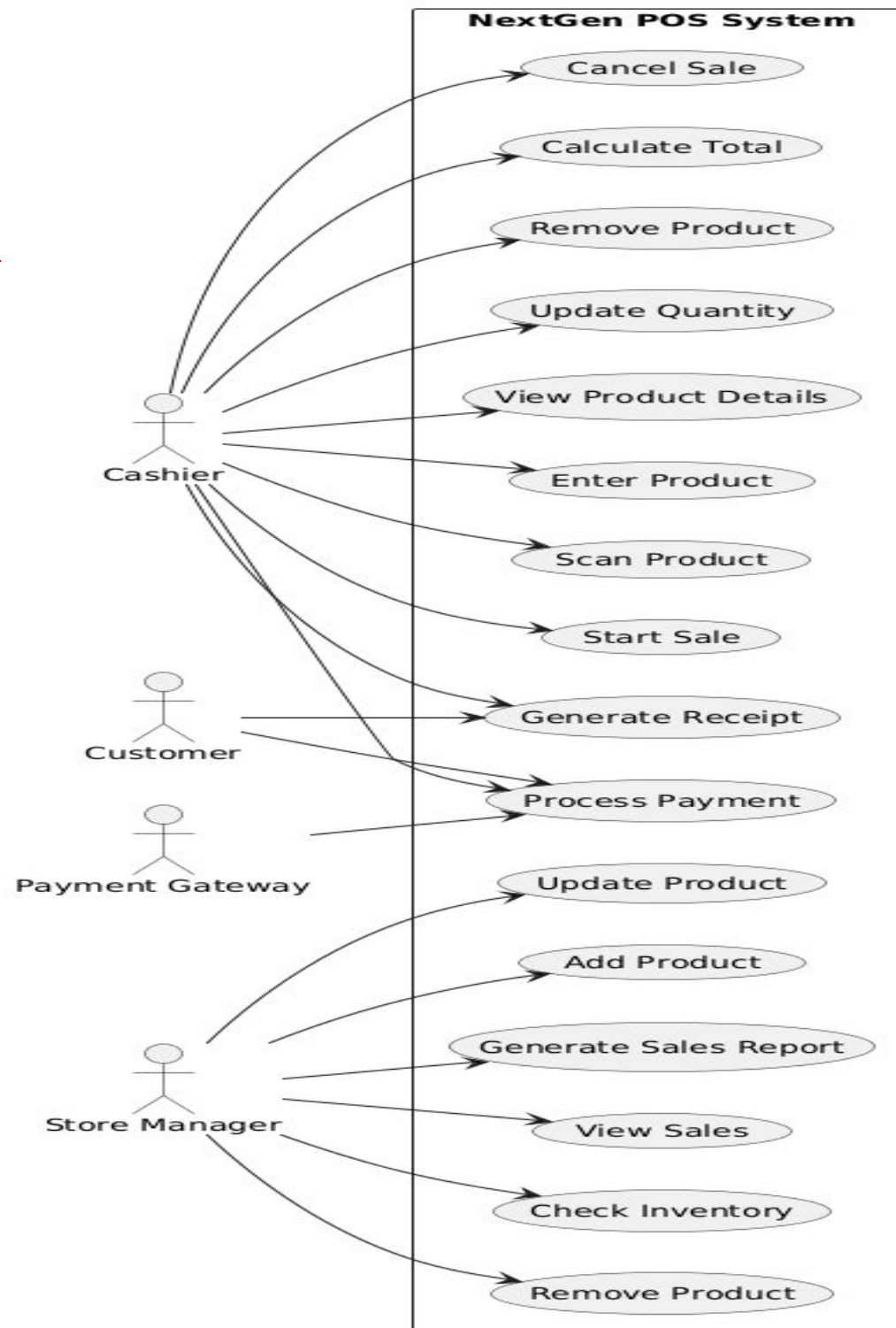
Payment Gateway -

The Payment Gateway participates in:

- Process Digital Payment
- Confirm Payment

Real-World Shopping Example

- Customer brings 2 soaps, 1 shampoo
- Shampoo removed by request
- System calculates total
- Customer pays → Change given
- Receipt printed
- Inventory updated



System Benefits

- Fast & accurate billing
- Auto-updated inventory
- Flexible payment options
- Sales analytics
- Eco-friendly digital receipts

THANK YOU